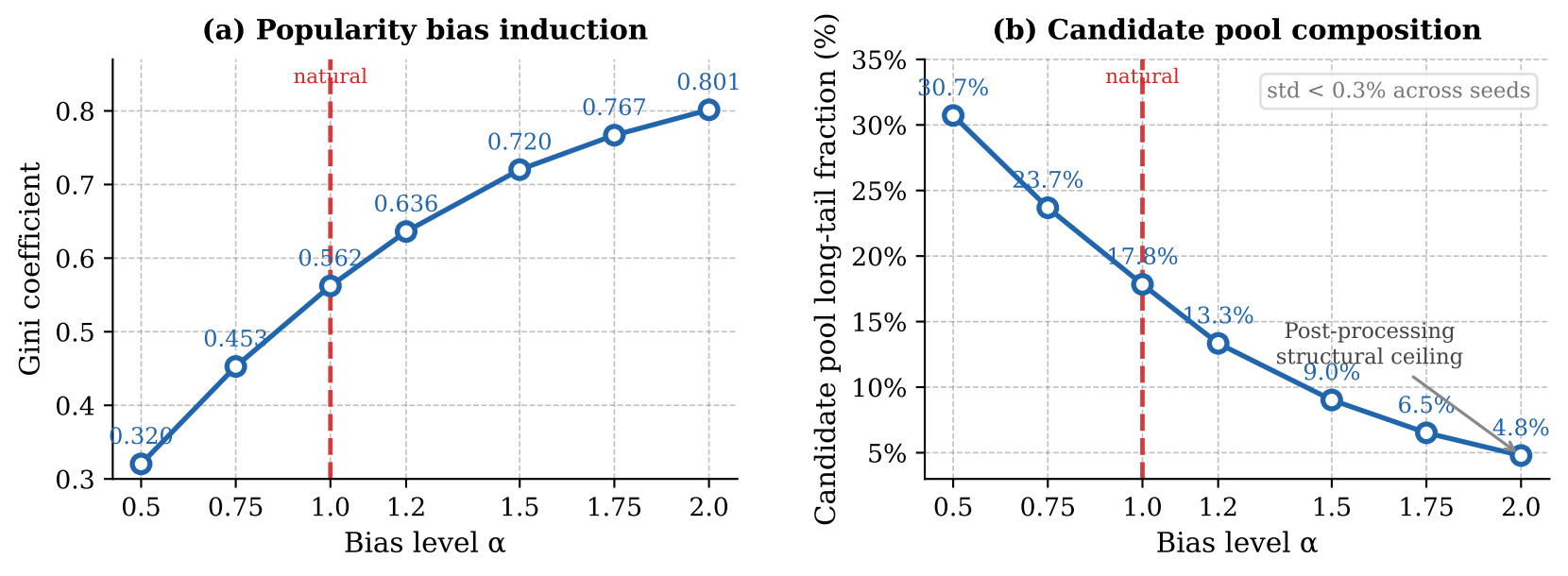
(a) Popularity bias induction
natural
0.801
0.767
0.720
0.636
0.562
0.453
0.320
Gini coefficient
0.8
0.7
0.6
0.5
0.4
0.3
0.5
0.75
1.0
1.2
1.5
1.75
2.0
Bias level α
(b) Candidate pool composition
natural
std < 0.3% across seeds
30.7%
23.7%
17.8%
13.3%
Post-processing
structural ceiling
9.0%
6.5%
4.8%
Candidate pool long-tail fraction (%)
35%
30%
25%
20%
15%
10%
5%
0.5
0.75
1.0
1.2
1.5
1.75
2.0
Bias level α